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Growing green? Sectoral-based prediction of GHG emission in Pakistan: a novel NDGM and doubling time model approach

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Abstract
Greenhouse gases (GHGs) are one of the leading causes of global warming. Therefore, accuracy estimates for greenhouse gases (GHGs) emissions are a key element in defining the best strategies for reducing GHG emissions from various source sectors of the economy. In the present study, an initial attempt has been made to estimate and forecast the GHG emissions in Pakistan from five major sectors, such as energy, industrial, agriculture, waste, and land-use change and forestry. The data were taken from the official website of the Pakistan climate database from 1990 to 2016. We employed advanced mathematical modeling, namely a non-homogenous discrete grey model (NDGM), to predict sector-wise GHGs emissions. Moreover, the present study is a milestone in the GHGs growth analysis by utilizing the synthetic relative growth rate (SRGR) and synthetic doubling time model (SDTM). The results reveal that the industry and land-use change and forestry contribute more in terms of increasing GHGs emissions till 2024, whereas agriculture and waste required comparatively less time to reduce GHGs emissions double in number among five sectors. All five sectors show an increasing trend in forecasting GHGs emissions between 1990 and 2016. However, the results indicate that land-use change and forestry and industrial sectors are more likely to be a reason for the increase in GHGs emissions in the future, followed by the agriculture, energy, and waste sectors. The land-use change and forestry was found prone to increase emission in the future, and the doubling time (D_t) suggests less time expected to reduce GHGs. Finally, the study has suggested some policies for the policymakers, government, and decision-makers to reduce GHGs emissions and achieve sustainable development.

Keywords Sustainable development · GHGs · Emission · Sectors · Pakistan · Green

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1 Introduction

Global climate change is an important issue faced globally by policymakers. The leading cause is the intensity of carbon dioxide (CO_2) and other gases (i.e., CH_4, HFC, PFC, N_2O, and SF_6) emissions in the environment, as fossil fuels are being used since the industrialization of the eighteenth century and as a result of agricultural practices related to increase in the global population (Seinfeld and Pandis 2006). The atmospheric carbon concentration was around 40% higher in 2011 than its pre-industrial level (Le Quéré et al. 2016). Moreover, global warming is a driving factor for climatic change, and the United Nations Framework Convention Climate Change (UNFCCC) is struggling to ensure that the average global temperature does not rise beyond 2 °C above the pre-industrial level so that uncontrollable adverse effects of climate change could be avoided. In this connection, the Kyoto protocol was signed in 1997 and entered into force in 2005, with binding commitments on the emissions of greenhouse gas (GHG) by the industrialized countries (Grunewald and Martinez-Zarzoso 2016). European Community and the thirty-seven industrialized nations committed themselves to the reduction of GHG emissions to an average of 5% over the 1990 level in the first commitment period (2008–2012). These goals were, however, not achieved successfully (Aichele and Felbermayr 2013). The UNFCCC recently signed an international agreement, called the Paris Climate Agreement, at its 21st Conference of the Parties (COP-21), that promised to main global temperature under 2 °C and attempted to restrict it 1.5 °C (Schleussner et al. 2016). This is certainly a breakthrough, and it is commendable that the agreement is ambitious, but the work is yet to start (Rogelj et al. 2016).

A key element of UNFCCC's effort to stabilize GHG elevated levels and prevent unsustainable levels of anthropogenic global warming is to provide an organized record of countries' GHG emissions, so all their trends over time can be monitored adequately (Ikram et al. 2019). According to UNFCCC Article 4 and 12, all developing countries which primarily come in non-Annex I Parties are required to assemble and submit an inventory of national-level GHG emissions to the Parties Conference following the policy guidance set out in the Annex Decision 17 (Janerio 1993).

Municipal solid waste (MSW) that produces a non-CO_2 GHG is considered to be the fourth major sector which contributes to global warming and climate change by emissions and adding around 5.5 to 6.4 percent to annual global emissions of methane 550 Tg (Ngwabie et al. 2019; Zhang et al. 2019). Methane is generated by the excretion of organic substances either by dropping these municipal solids in landfills or handling and procedure of municipal sewage sludge (Qu et al. 2019; Tzemi and Breen 2019). The landfills, which are the third major anthropological primary cause of methane emissions, come through agricultural activities and enteric fermentation (Chidambarampadmavathy et al. 2017; Ikram et al. 2020b). It accounts for 11 percent of the world's bio-gas emissions of methane in 50–55% by volume. It is expected that these emissions from landfills will increase by 816 Mt CO_2-eq by 2020. To make a significant contribution to this goal, the Government of Pakistan (GOP) has constructed and submitted its GHG emission inventory covering various socio-economic sectors for the year 1994 in compliance with the Intergovernmental Panel on Climate Change (IPCC) policies and send it to UNFCCC in 2003 together with its Initial National Communication (Distr 2003; Sun et al. 2020).

In the current study, the GHG emissions which include CH_4, CO_2, and N_2O are assessed based on the environmental activities in different sectors (land-use change and forestry (LUCF), industry, agriculture, waste, energy for the year 1990 to 2015 at the national level. All these GHG emission estimates are collectively presented as carbon dioxide CO_2

equivalents. As a result, the global warming potential (GWP) of N_2O, CH_4, and CO_2 was taken as 310, 21, and 1 based on the duration of 100 years set out in the second assessment report of IPCC, following the underlined standards of Annex Decision 17 (Michaelowa and Michaelowa 2015). The sectoral breakthrough of GHGs emission in Pakistan is presented in Fig. 1.

Pakistan is a sparsely populated country with over 199 million inhabitants in 2014, making it the sixth most populated country in the world (Pakistan Bureau of Statistics 2014). According to the Intergovernmental Panel on Climate Change (IPCC), Fifth Assessment Report (AR5) for the Asia region states that the susceptibility to climate change threats in agricultural-dependent economies, i.e., Pakistan stems from their distinctive geography, demographic changes, socio-economic factors, and lack of adaptive potential, which together determine the vulnerability profile of perpetuation. Population growth and gross domestic product (GDP) have put considerable strain on natural resources while simultaneously increasing emissions of GHGs, air pollution, and health impacts (Baig and Baig 2014; Uddin and Wadud 2014; Climatelinks 2019). The sectoral breakthrough of GHG emission of Pakistan is presented in Fig. 1. Ironically, none of Pakistan's cities has an effective and designed waste management system. The Pakistan Environmental Protection Act (PEPA) has been ineffectively applied, and its rules do not seem to be a priority for solidified waste management. Inadequate waste management practices have adverse health and environmental effects and make the situation especially, in urban areas more complex.

This study aims to forecast the GHGs emission from 1990 to 2015 for five highly GHG emitted sectors which include energy, Industrial, agriculture, waste, and Land use change and forestry sectors by applying a non-homogenous discrete grey model (NDGM). The data used in this study were gathered from Pakistan's official climate database website for the period 1990 to 2015. Further, novel synthetic relative growth rate (RGR) and synthetic

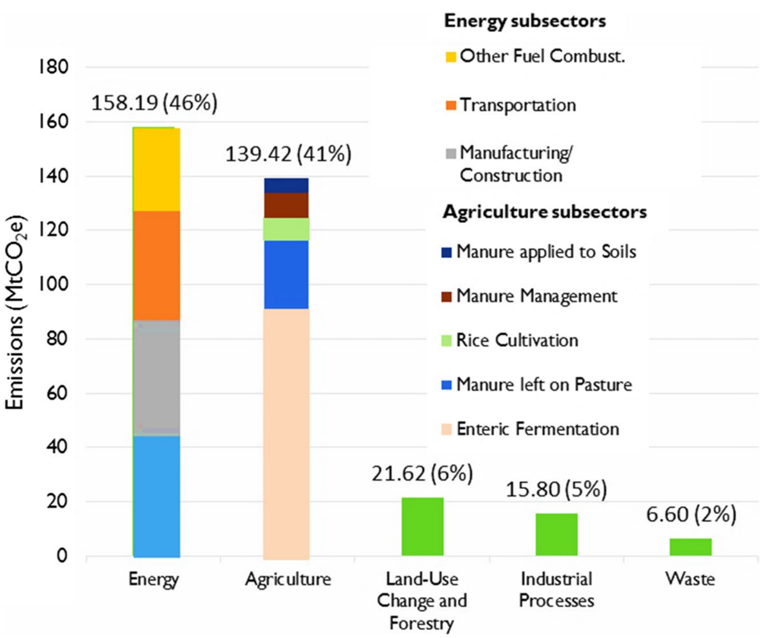

Fig. 1 Sectoral breakthrough of GHG emission in Pakistan (Sources: WRI CAIT 2.0, 2015; FAOSTAT)

doubling time (D_t) models were employed to undertake a comparative analysis of GHG emission relative growth rate among five sectors. Moreover, the MAPE% model was used to measure the level of accuracy for the proposed NDGM model. Finally, synthetic growth rate and synthetic doubling time models were employed to investigate the comparative analysis of five sectors in the long run regarding GHGs emissions. Hence, the present research is a pioneer study to forecast relative growth and required time to reduce the extent of GHG emission among five highly GHGs emitter sectors in Pakistan. Finally, the study proposed some suggestions for the government, policy, and decision-makers to take the appropriate measures to reduce GHGs emissions as well as to achieve sustainability.

The rest of the paper is follows as: Sect. 2 provides the literature review of GHG emission in Pakistan; Sect. 3 presents the data collection and research method used for analysis. Section 4 explains the results and discussion. Finally, the conclusion and policy implication for the research are mentioned in Sect. 5.

2 Literature review

Many researchers carried out different studies for GHG emission sector-wise, e.g., industry, agriculture, transportation, etc. (Khan and Siddiqui 2017; Ren et al. 2014; Jiang et al. 2019). In this research, the emissions from GHG are evaluated as a result of the environment activities carried out at the national level between the year 1990 and 2015 in various sectors such as agriculture, waste, industrial processes, energy, land-use change and forestry (Tariq et al. 2017; Gocht et al. 2016; Khan et al. 2019). It has emerged that, due to the rapid increase in developing countries, such as China, India and Korea, and the subsequent energy needs, particularly clear energy, like natural gas, useful information has clearly been given to decision-makers in advance. Again, this triggers a massive surge in research on this topic. For example, (Zheng et al. 2020) predicted Chinese natural gas consumption from 2019 to 2023 on the basis of an optimized non-homogeneous grey model (ONGM (1, 1)). Similarly, Gao et al. (2015) used four grey models to forecast projected Chinese Carbon emission from 2013 to 2015 and Şahin (2019) used a linear and nonlinear rolling grey metabolic model based on optimization to forecast Turkey's GHG emissions, including land use, land use change and forestry (LULUCF), except LULU and the energy sector.

Since the industrial revolution in 1990, Pakistan's energy demands have been increasing exponentially every year. Over the last 19 years, primary energy consumption in Pakistan almost has risen to 67 Mtoe in 2013–2014 from an estimated 34 Mtoe of oil equivalent from 1994 to 1995 (Board et al. 2017). Although Pakistan is not predicted to play an important role in the anthropogenic climate change, the emissions from Pakistan are mainly based on the energy sector electricity, transport, and manufacturing, and one of the main sources contributing to air pollution and the GHG emissions (Purohit et al. 2019).

The energy sector is responsible for 46% of total Pakistani yearly GHG emissions, electricity consumption accounted for 26% of them, manufacturing for 25%, transportation for 23%, and other energy subsectors for 25% of the remaining emissions. Between 1990 and 2012, it expanded by 87 $MtCO_2e$, which accounted for 55% of greenhouse gas emissions. In 2008, Pakistan had 147.8 million tons of CO_2e equivalent emissions in total (Bakhshal Shah et al. 2018; Khan and Siddiqui 2017). The subsectors of electricity and transport have been the big catalyst of a growing energy sector with 30% and 27% of the total energy sector growth, respectively (Akram et al. 2019). In the electricity generation mix, the relative proportion of GHG-intensive fossil fuels (coal, oil, and natural gas) expanded from 54% in

1990 to 64% in 2012 (USAID 2016). The overall energy usage has also increased during this period, growing 31 billion kWh in 1990 to 77 billion kWh, from 60% of the population in 1990 to 94% in 2012, driven by improved electricity access. The total consumption of fossil fuel in Pakistan's energy sector was 47.96 Mtoe in 2012. Out of this, 14.19 and 13.26 Mtoe were used in the electricity generation sector and manufacturing sector, respectively, while 10.06, 1.24, and 6.46 Mtoe were for transportation, commercial/organization sector, and household sector, respectively (Stiebert 2016; Qureshi et al. 2016). Industrial process activity data are obtained from the Pakistan Economic Survey (Pakistan Bureau of Statistics 2014). This comprises the production in Pakistan's chemical, mineral, and metal industries, while these mineral products include the processing of limestone paper, cement, ash, and dolomites, and the production of iron, urea, steel, and the chemical and metal industry provides ammonia. The total amount of soda ash, dolomite, and limestone was 0.17, 0.37, and 1.28 Tg, respectively, while the amount of cement production was nearly 29.56 Tera grams (Tg) in 2012. Furthermore, the amount of urea produced by the chemical industry was 4.47 Tg, and the quantity of iron and steel produced by the metal industry was 0.25 Tg (Akram et al. 2019).

For the agriculture sector in Pakistan, the activity data comprised livestock population (i.e., mules, sheep, dairy and non-dairy cattle, buffalos, poultry, horses, asses, camels, goats, etc.) for assessing the emissions of methane from the domestic livestock manure management and enteric fermentation in 2012. In the same way, yearly data on crop production for sugarcane, paddy, and wheat have been collected to estimate the methane emissions from the burning of the residue of crops. Furthermore, 41% of total GHG emissions in the agriculture sector are mainly from enteric fermentation (46%), while the other 18 percent appeared from synthetic fertilizers and 14 percent of it happened to come from manure left on the pasture. Also, data are assimilated to quantify nitrogen intake from agricultural soil by application of synthesis fertilization, animal waste, and crop residues (Board et al. 2017; Akhmat et al. 2014).

The agriculture sector contributed 150 $MtCO_2e$ GHG emissions in Pakistan in 2014, which represented 45.10% of its total emissions excluding land-use change and forestry (333 $MtCO_2$), and 41.53% including LUCF 362 $MtCO_2$. Most terrestrial biospheres have been converted into anthropic biomes by human populations and their land use. Around 4 percent of Pakistan's total land area covers forests, of which 5 percent is protected (Akhmat et al. 2014; Zaman 2012 Bushra et al. 2018). This transition has resulted in the formation of a variety of new environmental trends and processes that has been influential for more than 8000 years (Canadell et al. 2007; Johnson et al. 2011; Colomb et al. 2014). Also, there are indirect impacts of other human activities on land cover, for example, forests and lakes that have been impacted by acid rain from the combustion of fossil fuel and crops in the area of cities are affected by tropospheric ozone caused by toxic fumes of vehicles (Schleussner et al. 2016; Meyer 1999). Moreover, these interventions in humans, largely caused by socio-economic considerations, often lead to changes in unbuilt and built-up land, given physical restrictions (Liu et al. 2012). Changes in land use, which include land conversion from one kind to another and land cover alteration by land use management, have affected a significant proportion of the land surface of the earth. The aim is to meet the essential needs of human beings from natural resources (Turner et al. 2007; Franco Solís and De Miguel Vélez 2018).

The changes worldwide to forests, agricultural land, rivers, and streams and air are caused by the fact that more than six billion people need food, water, fiber, and shelter. The last decade has seen the growth of global agricultural fields, plantations, urban areas, and pastures.

This expansion is followed by significant growth in the consumption of energy, fertilizer, and water along with major biodiversity losses (Lambin et al. 2001; Zaks et al. 2009).

In Pakistan, nearly 20 Tg of solid waste is produced yearly, with a growth rate of approximately 2.4 percent annually (Fulton et al. 2017). These data consist of municipal solid waste which encompasses the yearly local solid waste disposal locations for the assessment of methane emissions produced from the land waste disposal (Khan and Ghouri 2011; Kawai and Tasaki 2016). Javed and Khan (2012), examined land usage/land cover changes activities between 2001 to 2010; they found that there has been a significant reduction of forested area, agricultural soil, and water resources, but mostly anthropogenic activities have expanded due to settlements, uncultivated land, and wastelands (Akram et al. 2018; Rothwell et al. 2016). Changes in land use may influence each other. Many environmental impacts from land-use change represent collaborative effects as a result of a diverse change in land use. Deforestation, for instance, resulted in freshwater habitats being destroyed as rivers stilted. Likewise, as a carbon sink and source, the position of the Asian forest varies from year to year or from place to place because of its interconnected impacts between deforestation, reforestation, and afforestation. Consequently, the associations between the various land needs along their changing paths pose 53 obstacles to understand better the issue of land-use change and forestry. Land-use and ecosystem changes and their impact on global climate change and sustainability are leading research concerns in environmental sciences for researchers and humans also (Lambin et al. 2001; Balbus et al. 2007; Turner et al. 2007).

In this study, we employed the Grey forecasting model, namely NDGM to estimate the GHG emission in five major sectors of Pakistan. Also, this study is a milestone in the GHGs growth analysis by utilizing the synthetic relative growth rate (SRGR) and synthetic doubling time model (SDTM) that will help to analyze the comparatively less time to reduce GHGs emissions double in number among five sectors.

3 Data collection and method

3.1 Data

Pakistan's GHG emission dataset was abstracted from Pakistan's official climate website for the period 1990–2016. The five significant sectors, i.e., the energy, agriculture, industrial, land-use change and forestry, and waste sectors of Pakistan, which are the main cause of GHGs emissions, have been selected for our research. Further, a Grey system software (version 8.0) was used to forecast the emission of GHGs for each sector from 2016 to 2024 by a non-homogeneous discrete grey model (NDGM). However, MS EXCEL and MATLAB were also utilized to solve NDGM. The current analysis and modeling methodology were used for the first time in the forecasting study to assess the sector's wise impact on GHG emissions. The structure of forecasting the GHG emission is operationalized in this study presented in Fig. 2.

3.2 Grey forecasting model

3.2.1 Non-homogeneous discrete grey model (NDGM)

The NDGM system is designed based on non-homogenous exponential growth approximation law in order to comply with the assumptions of an actual data sequence. Xie et al. (2013)

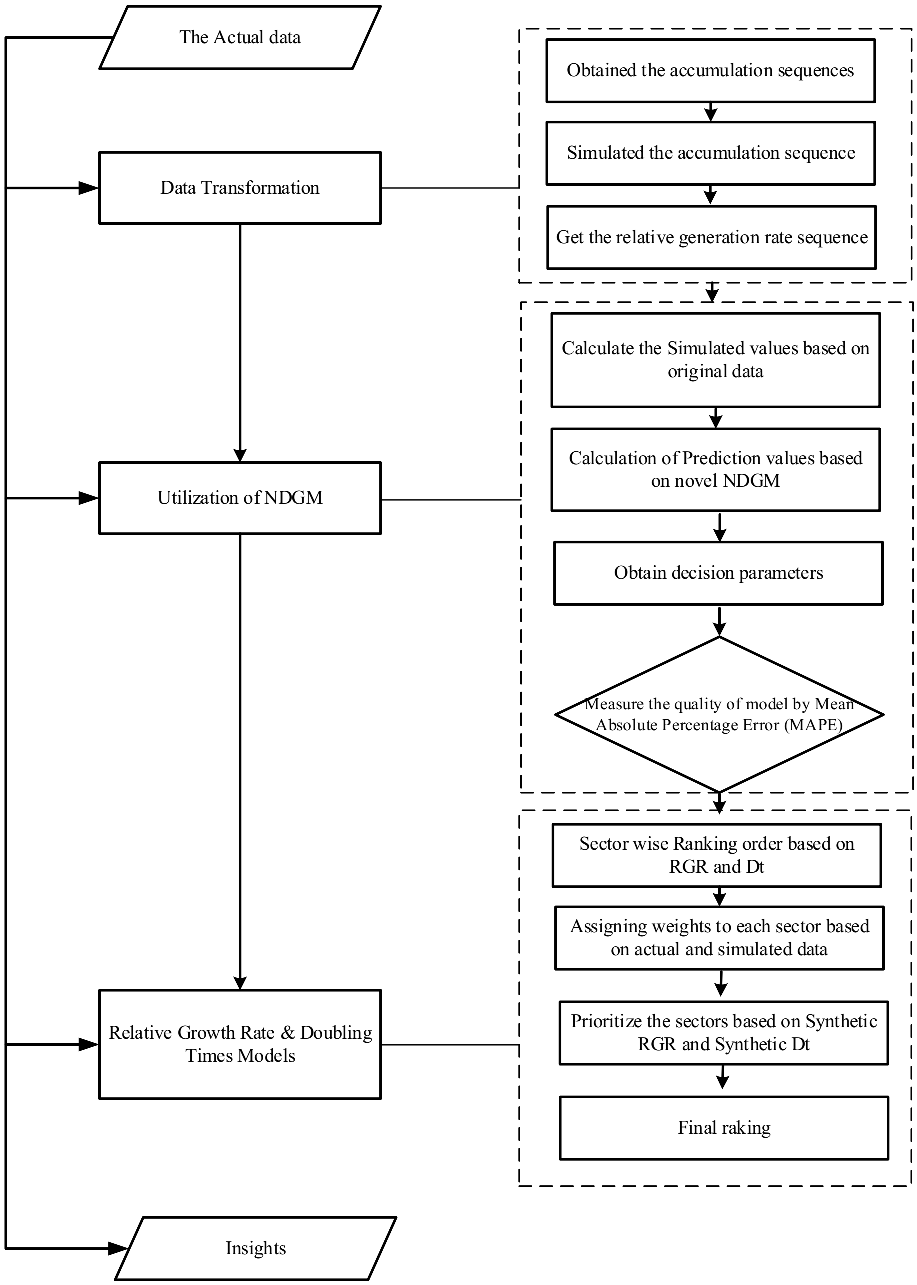

Fig. 2 Structure of forecasting operationalized in this study

recommended that the real data sequence competes with a homogeneous pattern such as GM (1, 1). The accuracy of the NDGM model is significantly improved over the other grey models as far as the mean sequence value and the value of the intervals are concerned. NDGM model has been used in various disciplines; for example, predicting Turkey's electricity consumption and analyzing the NDGM is the best fit and predicting more accurately than other grey predicting models. Ikram et al. (2020a) predicted the future trend of ISO 9001 certification of by utilizing the NDGM model. Moreover, Duan et al. (2018) predicted the consumption of crude oil in China and analyzed that the NDGM showed superior performance.

In the model of NDGM,$x^{(0)}$ denotes the original data sequence and $x^{(1)}$ represents the accumulated sequence of data. So, we write it as follows:

$$\hat{x}^{(1)}(L+1) = \beta_1 \hat{x}^{(1)}(L) + \beta_2 \cdot L + \beta_3 \tag{1}$$

$$\hat{x}^{(1)}(1) = x^{(1)}(1) + \beta_4 \tag{2}$$

$\hat{x}^{(1)}(L)$, represents the predicted value $x^{(1)}$ along with their parameters β_1, β_2, β_3, and β_4. So, the above equation can be written in matrix form as follows, where $L=1$, 2, and 3… n-1

$$\begin{bmatrix} x^{(1)}(2) \\ x^{(1)}(3) \\ \vdots \\ x^{(1)}(n) \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \begin{bmatrix} x^{(1)}(1) & 1 & 1 \\ x^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ x^{(1)}(n-1) & L-1 & 1 \end{bmatrix}$$

The input data in a single case indicate a constant sequence for the NDGM parameters β_1, β_2, β_3, and β_4, so by applying this relation in Eq. (3), we get it as:

$$\hat{\beta} = \left(B^{\mathrm{T}}B\right)^{-1} B^{\mathrm{T}} Y = \left[\beta_1, \beta_2, \beta_3\right]^{\mathrm{T}} \tag{3}$$

whereas,

$$B = \begin{bmatrix} x^{(1)}(1) & 1 & 1 \\ x^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ x^{(1)}(n-1) & L-1 & 1 \end{bmatrix};\ Y = \begin{bmatrix} x^{(1)}(2) \\ x^{(1)}(3) \\ \vdots \\ x^{(1)}(n) \end{bmatrix},$$

Further, to calculate β_4 the following formula is being used where we minimize the sum of square error as:

$$\beta_4 = \frac{\sum_{L=1}^{n-1}\left[x^{(1)}(L+1) - \beta_1^L x^{(1)}(1) - \beta_2 \sum_{j=1}^{L} j\beta_1^{L-j} - \frac{1-\beta_1^L}{1-\beta_1} . \beta_3\right] . \beta_1^L}{1 - \sum_{L=1}^{n-1}\left(\beta_1^L\right)^2} \tag{4}$$

$$\hat{x}^{(1)}(L+1) = \beta_1^L \overset{(1)}{x}(1) + \beta_2 \sum_{j=1}^{L} j\beta_1^{L-j} + \frac{1-\beta_1^L}{1-\beta_1} \cdot \beta_3; \quad L = 1, 2, \ldots, n-1 \tag{5}$$

For the additional knowledge of the NDGM model, its parameter and properties (Liu and Forrest 2010) are suggested (Liu et al. 2017).

3.2.2 Performance evaluation approach of NDGM

To assess the accuracy of the NDGM model, we used the MAPE. The formulae for mean absolute percentage error (MAPE)% computation is as follows:

$$\text{MAPE}(\%) = \frac{1}{n}\sum_{L=1}^{n}\left|\frac{x^{(0)}(L) - \hat{x}^{(0)}(L)}{x^{(0)}(L)}\right| \times 100\% \tag{6}$$

where $x^{(0)}(L)$ represents the actual values of the data. While $\hat{x}^{(0)}(L)$ denotes the forecasted data.

3.2.3 The growth of GHSs emissions and doubling time analysis

There is no model of GHG emissions available to monitor the rate of growth. RGR model was used in this way to analyze the sector's wise relative growth of GHG emissions. To reach this objective, the literature proposes the use of relative growth rate and doubling time models that have been used to analyze the ISO 14001 certification and research output growth over time (Ikram et al. 2019; Javed and Liu 2018).

Using the NDGM method, two parameters (D_t and RGR) were used to forecast GHGs emissions in five major sectors of Pakistan. The RGR formula is given by,

$$\text{RGR} = \left(\ln L_2 - \ln L_1\right)/\left(t_2 - t_1\right) \tag{7}$$

where L_2 depicts cumulative GHG emissions in the year t_2 and L_1 signifies cumulative GHG emissions in the year t_1. In the analysis, as $t_2 - t_1$ is one year, Eq. (7) can be interpreted as Eq. (8).

$$\text{RGR} = \ln\left(L_2/L_1\right) \tag{8}$$

And doubling time D_t depicts the time required to reduce GHGs emissions double in numbers for specified RGR. So, the D_t is presented as:

$$D_t = \left(t_2 - t_1\right)\ln\left[2/\left(\ln L_2 - \ln L_1\right)\right] \tag{9}$$

We can also rewrite this equation as follows:

$$D_t = \ln\left[2/\left(\ln L_2 - \ln L_1\right)\right] \tag{10}$$

If the relative growth rate (RGR) and doubling time (D_t) give a different pattern for actual and simulated data, in any case, the synthetic relative growth rate (RGR_{syn}) and synthetic doubling time (D_{syn}) models have been introduced to jump into the conclusion for further accuracy. The synthetic relative growth rate (RGR_{syn}) model is characterized in the following equation as:

$$D_{\text{syn}} = \theta\left(D_a\right) + (1-\theta)\left(D_f\right) \tag{11}$$

whereas RGR_a is a relative growth rate of actual values of data, while RGR_f depicts forecasted values of the relative growth rate. However, θ represents the relative weights coefficient and mostly it is considered to be 0.5.

The synthetic doubling time (D_{syn}) model is shown in Eq. (12)

$$D_{\text{syn}} = \theta\left(D_a\right) + (1-\theta)\left(D_f\right) \tag{12}$$

where D_a signifies the doubling time acquired from actual values, whereas RGR_f represents the relative growth rate of forecasted values.

4 Results

We employed the NDGM model to forecast the GHGs emission in five different sectors of Pakistan, i.e., Energy, agriculture, industrial sector, Land use change and forestry, and waste. The data utilized in this study are from 1990 to 2015. The simulated values calculated from the NDGM model for the GHG emission are given in Tables 1, 2, 3, 4, and 5. The results for the energy sector of Pakistan are displayed in Table 1. To check the accuracy of the proposed model, the mean percentage absolute error (MAPE) % has been utilized. The findings estimated through MAPE% show the level of effectiveness with the value 97.77% which demonstrates that the NDGM is a best-fit grey model to forecast the emission of GHGs.

For the case of energy sector GHGs emission in Pakistan, the parameters of NDGM are $\beta_1 = 0.967807$, $\beta_2 = 8{,}005{,}234.5$, $\beta_3 = 62{,}010{,}330.4$ and $\beta_4 = 86{,}968.41$. Additionally, the MAPE % for the energy sector of Pakistan is 97.42% which shows the accuracy level of the model. To get a clearer view, the actual data and the simulated data are estimated through NDGM from the year 1990 to 2024 against the total GHGs emission in quadrillions British units. According to simulated values of GHGs in energy sector computed by NDGM, a inclined trend from 1990 to 2014 is recorded which started from 68.34 million ($MtCO_2e$) to 163.60 million ($MtCO_2e$), while the predicted values of energy sector in Pakistan from 2015 to 2024 are 166.25, 168.50, 171.50, 173.90, 176.30, 178.80, 180.50, 183.10, 185.70, and 187.10 million ($MtCO_2e$), respectively.

Energy plays an important role in economic development and high consumption of energy has become a benchmark of living standard in the world. Whereas rapidly increase of energy consumption is positive indicator of economic growth, however, it also causes increase in emission of GHGs and climate vulnerability. In the past decade, energy sector has reported the high emitted sector of GHG in Pakistan and CO_2 emission are the most hazardous among others in GHGs due to consumption of fossil fuels (Ikram et al. 2020a). Previous studies have investigated the effect of energy consumption on environment by using statistical models and reported that the energy sector is high contributor in emission of CO_2 which make the climate vulnerable (Lin and Raza 2019; Rehman et al. 2020). Similarly, another study conducted by (Sun et al. 2020) endorsed our finding that the energy sector contributed about 50% of the total GHG emission in Pakistan. To fulfill the energy demand of Pakistan, the 99% of energy is produced by conventional methods, whereas less than 1% of energy demands is met by renewable methods (Kamran 2018).

Table 2 depicts the findings from the industrial sector of Pakistan with the MAPE of 98.50% which shows the level of accuracy. The calculated parameters for the NDGM forecasting equation are given as: $\beta_1 = 1.057032$, $\beta_2 = 223{,}823.87$, $\beta_3 = 2{,}029{,}683.7$, and $\beta_4 = 283{,}148.2$.

In Table 2, GHGs emissions of industrial sector have shown the increasing trend from 1990 to 2014 by utilizing the NDGM prediction model. The simulated values start 3.72 million ($MtCO_2e$) and reached 196.63 million ($MtCO_2e$). But original values of GHGs in the industrial sector are little different than simulated values which can be noted in the NDGM column of the table, whereas the forecasted values for the industrial sector in Pakistan have been calculated for the year 2015 to 2024. These predicted values are elevating from 20.27 million ($MtCO_2e$) and reached a level of 36.38 million ($MtCO_2e$).

It has reported that 59 billion bricks are annually produced by using various types of fuels such as coal, agriculture waste, plastic, rubber tires, rice husk and different types of industrial waste which are emitting hazardous and toxic gases and ultimately cause

Table 1 Forecasting the energy sector GHGs emission in Pakistan

Years	Energy	NDGM	Cumulative	RGR	Mean RGR	D_t	Mean D_t
1990	68,335,925.3	68,335,925.34	68,335,925.34				
1991	67,873,281.7	67,812,794.16	136,209,207	0.690	0.157	1.065	2.804
1992	74,864,784.7	73,634,899.2	211,073,991.7	0.438		1.519	
1993	81,866,287.7	79,269,570.44	292,940,279.3	0.328		1.809	
1994	84,847,790.7	84,722,842.05	377,788,070	0.254		2.062	
1995	91,529,293.7	90,000,553.89	469,317,363.7	0.217		2.221	
1996	96,045,113.7	95,108,357.86	565,362,477.4	0.186		2.374	
1997	99,950,933.8	100,051,723.9	665,313,411.2	0.163		2.508	
1998	101,736,754	104,835,945.8	767,050,165.1	0.142		2.643	
1999	110,502,574	109,466,146.9	877,552,739.1	0.135		2.699	
2000	110,348,394	113,947,285.8	987,901,133.2	0.118		2.826	
2001	112,018,946	118,284,161.3	1,099,920,080	0.107		2.924	
2002	115,039,499	122,481,417.7	1,214,959,578	0.099		3.001	
2003	118,550,051	126,543,549.8	1,333,509,629	0.093		3.067	
2004	131,510,603	130,474,907.8	1,465,020,231	0.094		3.057	
2005	134,291,155	134,279,701.7	1,599,311,386	0.088		3.127	
2006	144,311,698	137,962,006.1	1,743,623,084	0.086		3.142	
2007	156,302,240	141,525,764.3	1,899,925,325	0.086		3.148	
2008	150,982,783	144,974,792.7	2,050,908,108	0.076		3.264	
2009	154,773,326	148,312,784.9	2,205,681,433	0.073		3.314	
2010	151,203,868	151,543,315.6	2,356,885,302	0.066		3.407	
2011	152,845,089	154,669,844.3	2,509,730,391	0.063		3.460	
2012	155,000,429	157,695,719.1	2,664,730,820	0.060		3.508	
2013	156,291,650	160,624,180.5	2,821,022,470	0.057		3.558	
2014	158,962,870	163,458,364.6	2,979,985,340	0.055		3.597	
2015		166,201,306.5	166,201,306.5				
2016		168,855,943.5	335,057,249.9	0.701		1.048	
2017		171,425,118.5	506,482,368.4	0.413		1.577	
2018		173,911,582.9	680,393,951.3	0.295		1.913	
2019		176,317,999.3	856,711,950.6	0.230	0.263	2.161	2.211
2020		178,646,944.8	1,035,358,895	0.189		2.357	
2021		180,900,913.5	1,216,259,809	0.161		2.519	
2022		183,082,319.1	1,399,342,128	0.140		2.658	
2023		185,193,497.7	1,584,535,626	0.124		2.778	
2024		187,236,710.1	1,771,772,336	0.112		2.885	

air pollution, whereas the Pakistan fertilizer demand has increased for last 30 years and production capacity has reached 10 million metric tons yearly (69% for Urea and 31% for DAP and potash), and natural gas is used as fuel for production of fertilizer. Moreover, the cement industry of Pakistan produced 4100 thousands of tons yearly and contributes considerably to Gross Domestic Production (GDP). The domestic and imported coal is the major fuel consumed in production of cement in Pakistan. According to World Steel Association (WSA), the total Pakistan's steel consumption in 2018 was recorded 4.7 million

Table 2 Forecasting the industrial sector GHGs emission in Pakistan

Years	Industry	NDGM	Cumulative	RGR	Mean RGR	D_t	Mean D_t
1990	3,798,821.7	3,798,821.72	3,798,821.72				
1991	3,938,114.9	2,486,309.25	7,736,936.64	0.711	0.170	1.034	2.638
1992	3,956,496.1	2,851,931.56	11,693,432.76	0.413		1.577	
1993	4,224,029.3	3,238,405.94	15,917,462.09	0.308		1.870	
1994	4,117,834.5	3,646,921.6	20,035,296.62	0.230		2.162	
1995	4,363,383.7	4,078,735.61	24,398,680.35	0.197		2.318	
1996	4,549,678.8	4,535,176.71	28,948,359.19	0.171		2.459	
1997	4,633,381.9	5,017,649.41	33,581,741.13	0.148		2.601	
1998	4,618,157	5,527,638.36	38,199,898.17	0.129		2.742	
1999	4,998,644.2	6,066,712.84	43,198,542.32	0.123		2.789	
2000	5,177,611.3	6,636,531.65	48,376,153.57	0.113		2.872	
2001	5,817,989.5	7,238,848.2	54,194,143.05	0.114		2.869	
2002	5,908,767.71	7,875,515.892	60,102,910.76	0.103		2.961	
2003	6,996,153.93	8,548,493.82	67,099,064.69	0.110		2.899	
2004	8,083,540.16	9,259,852.823	75,182,604.85	0.114		2.867	
2005	9,170,926.39	10,011,781.84	84,353,531.24	0.115		2.855	
2006	11,059,791.7	10,806,594.64	95,413,322.96	0.123		2.787	
2007	13,663,137	11,646,736.98	109,076,460	0.134		2.704	
2008	16,251,826.4	12,534,794.05	125,328,286.4	0.139		2.667	
2009	15,869,011.7	13,473,498.53	141,197,298.1	0.119		2.820	
2010	15,988,165	14,465,738.93	157,185,463.1	0.107		2.926	
2011	15,578,622.6	15,514,568.47	172,764,085.7	0.095		3.052	
2012	16,165,688.3	16,623,214.56	188,929,774	0.089		3.107	
2013	16,752,753.9	17,795,088.61	205,682,527.9	0.085		3.159	
2014	17,339,819.5	19,033,796.63	223,022,347.4	0.081		3.207	
2015		20,343,150.27	20,343,150.27				
2016		21,727,178.58	42,070,328.85	0.727		1.013	
2017		23,190,140.38	65,260,469.23	0.439		1.516	
2018		24,736,537.37	89,997,006.6	0.321		1.828	
2019		26,371,128.01	116,368,134.6	0.257	0.290	2.052	2.079
2020		28,098,942.14	144,467,076.7	0.216		2.224	
2021		29,925,296.44	174,392,373.2	0.188		2.363	
2022		31,855,810.83	206,248,184	0.168		2.478	
2023		33,896,425.74	240,144,609.7	0.152		2.576	
2024		36,053,420.38	276,198,030.1	0.140		2.660	

tones and ranked on 32 number globally (WSA 2019). As per world bank statistics, in 2014 Pakistan contributed 23.84% CO_2 emissions from manufacturing industries and construction (% of total fuel combustion) (The World Bank 2020).

Table 3 shows the results of the agriculture sector with 97.69% MAPE accuracy level and the calculated parameters for NDGM forecasting equations are $\beta_1 = 1.034794$, $\beta_2 = -591{,}865.01$, $\beta_3 = 76{,}086{,}437.7$, and $\beta_4 = 37{,}308.19$.

Table 3 Forecasting the agricultural sector GHGs emission in Pakistan

Years	Agriculture	NDGM	Cumulative	RGR	Mean RGR	D_t	Mean D_t
1990	77,097,486	77,097,486	77,097,486				
1991	78,236,535	78,178,362.7	155,334,021.3	0.701	0.149	1.049	2.869
1992	80,371,503	80,306,597.16	235,705,524.6	0.417		1.568	
1993	82,680,670	82,508,880.36	318,386,194.6	0.301		1.895	
1994	84,416,060	84,787,788.72	402,802,254.7	0.235		2.141	
1995	87,871,568	87,145,988.29	490,673,823.1	0.197		2.316	
1996	90,620,650	89,586,237.9	581,294,472.8	0.169		2.468	
1997	93,126,814	92,111,392.35	674,421,286.9	0.149		2.600	
1998	95,110,789	94,724,405.78	769,532,076.2	0.132		2.719	
1999	98,121,172	97,428,335.12	867,653,248.2	0.120		2.813	
2000	99,801,592	100,226,343.6	967,454,840	0.109		2.911	
2001	1.01E+08	103,121,704.7	1,068,268,041	0.099		3.005	
2002	1.04E+08	106,117,805.5	1,172,034,975	0.093		3.072	
2003	1.07E+08	109,218,151.2	1,279,223,986	0.088		3.129	
2004	1.1E+08	112,426,368.8	1,389,442,187	0.083		3.186	
2005	1.15E+08	115,746,211.5	1,503,952,328	0.079		3.229	
2006	1.21E+08	119,181,563.2	1,625,350,477	0.078		3.249	
2007	1.23E+08	122,736,442.8	1,748,596,109	0.073		3.309	
2008	1.29E+08	126,415,009.2	1,877,530,752	0.071		3.336	
2009	1.35E+08	130,221,565.8	2,012,899,317	0.070		3.358	
2010	1.34E+08	134,160,565.8	2,146,416,164	0.064		3.439	
2011	1.39E+08	138,236,617.5	2,285,551,568	0.063		3.461	
2012	1.39E+08	142,454,489.3	2,424,926,742	0.059		3.520	
2013	1.47E+08	146,819,115.6	2,571,985,082	0.059		3.525	
2014	1.5E+08	151,335,602.7	2,722,325,853	0.057		3.561	
2015		156,009,234.1	156,009,234.1				
2016		160,845,477.6	316,854,711.7	0.709		1.038	
2017		165,849,991	482,704,702.7	0.421		1.558	
2018		171,028,628.9	653,733,331.6	0.303		1.886	
2019		176,387,449.8	830,120,781.4	0.239	0.272	2.125	2.164
2020		181,932,722.9	1,012,053,504	0.198		2.312	
2021		187,670,935.5	1,199,724,440	0.170		2.464	
2022		193,608,800.6	1,393,333,240	0.150		2.593	
2023		199,753,264.9	1,593,086,505	0.134		2.703	
2024		206,111,516.6	1,799,198,022	0.122		2.800	

In the same way, identical results were derived by the NDGM model given in Table 4. Likewise, the agriculture sector is the major contributor in GDP and has considered the backbone of economy development. The original data are starting from 77 million ($MtCO_2e$) to 151 million ($MtCO_2e$) from 1990 to 2014, whereas simulated values and original values are approximately the same. The increasing trend has been observed from 1990 to 2014, while forecasted values have been calculated for the year 2015 to

Table 4 Forecasting the waste GHGs emission in Pakistan

Years	Waste	NDGM	Cumulative	RGR	Mean RGR	D_t	Mean D_t
1990	4,932,198.4	4,932,198	4,932,198.36				
1991	5,007,346.8	4,987,324	9,939,545.16	0.70	0.14	1.05	2.96
1992	5,082,495.2	5,074,288	15,022,040.4	0.41		1.58	
1993	5,157,643.7	5,160,179	20,179,684.09	0.30		1.91	
1994	5,232,792.1	5,245,012	25,412,476.22	0.23		2.16	
1995	5,307,940.6	5,328,798	30,720,416.79	0.19		2.36	
1996	5,395,681.3	5,411,552	36,116,098.12	0.16		2.51	
1997	5,483,422.1	5,493,286	41,599,520.21	0.14		2.65	
1998	5,571,162.9	5,574,012	47,170,683.06	0.13		2.77	
1999	5,658,903.6	5,653,743	52,829,586.67	0.11		2.87	
2000	5,746,644.4	5,732,492	58,576,231.04	0.10		2.96	
2001	5,820,334.4	5,810,269	64,396,565.42	0.09		3.05	
2002	5,894,024.4	5,887,088	70,290,589.82	0.09		3.13	
2003	5,967,714.41	5,962,959	76,258,304.23	0.08		3.20	
2004	6,041,404.43	6,037,896	82,299,708.66	0.08		3.27	
2005	6,115,094.44	6,111,909	88,414,803.1	0.07		3.33	
2006	6,185,422.97	6,185,009	94,600,226.07	0.07		3.39	
2007	6,255,751.5	6,257,208	100,855,977.6	0.06		3.44	
2008	6,326,080.02	6,328,517	107,182,057.6	0.06		3.49	
2009	6,396,408.54	6,398,947	113,578,466.1	0.06		3.54	
2010	6,466,737.07	6,468,509	120,045,203.2	0.06		3.59	
2011	6,534,907.44	6,537,214	126,580,110.6	0.05		3.63	
2012	6,603,077.81	6,605,071	133,183,188.5	0.05		3.67	
2013	6,671,248.19	6,672,092	139,854,436.6	0.05		3.71	
2014	6,739,418.56	6,738,287	146,593,855.2	0.05		3.75	
2015		6,803,666	6,803,665.845				
2016		6,868,239	13,671,904.56	0.70		1.05	
2017		6,932,016	20,603,920.15	0.41		1.58	
2018		6,995,006	27,598,926.41	0.29		1.92	
2019		7,057,220	34,656,146.85	0.23		2.17	
2020		7,118,668	41,774,814.54	0.19	0.26	2.37	2.22
2021		7,179,357	48,954,172	0.16		2.53	
2022		7,239,299	56,193,471.1	0.14		2.67	
2023		7,298,502	63,491,972.93	0.12		2.80	
2024		7,356,975	70,848,947.67	0.11		2.90	

2024 under the NDGM grey model which started about 156.1 million ($MtCO_2e$) and reached the level of approximately to 206.6 million ($MtCO_2e$).

A major part of Pakistan's economy relies on agriculture. 21% of GDP comes from agriculture and livestock, whereas 51% from the labor force and contributed to foreign reserve (Rehman et al. 2017). In the last few decades, livestock industry has become major portion of agriculture sector due to high population growth rate, improving the living standard, and high food demand. Meanwhile, the emission level of GHGs in agriculture sector has risen

Table 5 Forecasting the land-use change and forestry GHGs emission in Pakistan

Years	LUCF	NDGM	Cumulative	RGR	Mean RGR	D_t	Mean D_t
1990	21,635,333.3	21,635,333	21,635,333.3				
1991	21,635,333.3	20,941,783	42,577,116.7	0.68	0.14	1.08	2.99
1992	21,635,333.3	20,948,998	63,526,114.63	0.40		1.61	
1993	21,635,333.3	20,958,299	84,484,413.33	0.29		1.95	
1994	21,635,333.3	20,970,289	105,454,702.3	0.22		2.20	
1995	21,635,333.3	20,985,747	126,440,448.9	0.18		2.40	
1996	21,635,333.3	21,005,674	147,446,123	0.15		2.57	
1997	21,633,333.3	21,031,364	168,477,487	0.13		2.71	
1998	21,639,333.3	21,064,483	189,541,969.9	0.12		2.83	
1999	21,637,333.3	21,107,179	210,649,148.5	0.11		2.94	
2000	21,637,333.3	21,162,221	231,811,369.5	0.10		3.04	
2001	20,533,333.3	21,233,180	253,044,549.5	0.09		3.13	
2002	20,535,333.3	21,324,658	274,369,207.9	0.08		3.21	
2003	20,533,333.3	21,442,590	295,811,797.8	0.08		3.28	
2004	20,535,333.3	21,594,624	317,406,421.7	0.07		3.35	
2005	20,539,333.3	21,790,622	339,197,043.7	0.07		3.41	
2006	22,005,000	22,043,297	361,240,341	0.06		3.46	
2007	22,002,000	22,369,039	383,609,380.4	0.06		3.51	
2008	22,000,000	22,788,977	406,398,357.6	0.06		3.55	
2009	22,000,000	23,330,349	429,728,707.1	0.06		3.58	
2010	22,001,000	24,028,272	453,756,978.8	0.05		3.60	
2011	28,600,000	24,928,014	478,684,992.7	0.05		3.62	
2012	28,604,000	26,087,937	504,772,929.5	0.05		3.63	
2013	28,600,000	27,583,278	532,356,207.1	0.05		3.63	
2014	28,601,000	29,511,030	561,867,236.9	0.05		3.61	
2015		31,996,235	31,996,235.1				
2016		35,200,093	67,196,328.58	0.74		0.99	
2017		39,330,420	106,526,748.3	0.46		1.47	
2018		44,655,123	151,181,871.1	0.35		1.74	
2019		51,519,584	202,701,455	0.29	0.34	1.92	1.87
2020		60,369,059	263,070,513.6	0.26		2.04	
2021		71,777,558	334,848,071.8	0.24		2.12	
2022		86,485,083	421,333,154.6	0.23		2.16	
2023		105,000,000	526,778,774	0.22		2.19	
2024		130,000,000	656,667,796.6	0.22		2.21	

simultaneously. Agriculture is the primary source of many developing countries' economies, the CO_2 emission is increasing drastically due to low production of crop land and will ultimately negative affect socio-economic development (Abas et al. 2017).

Likewise, Table 4 represents the findings for waste sector of Pakistan with the MAPE accuracy level of 96.60%. The estimated parameters of the NDGM forecasting equation for the case of waste sector GHGs emission are $\beta_1 = 0.987673$, $\beta_2 = 148{,}443.18$, $\beta_3 = 4{,}899{,}681.79$ and $\beta_4 = 18{,}648.22$.

On the other hand, the waste sector is producing comparatively lower GHG emission than energy, industrial and waste sectors. The simulated values are starting from 4 million ($MtCO_2e$) and leveled up to 6 million ($MtCO_2e$) under NDGM. Table 4 presents the uprising trend of GHGs in agriculture sector from 1990 to 2024. The predicting values of agriculture sector in Pakistan from 2015 to 2024 are 6,803,666, 6,868,239, 6,932,016, 7,057,220, 7,179,357, 7,179,357, 7,239,299, 7,298,502, and 7,356,975 ($MtCO_2e$), respectively.

Solid waste management is a crucial issue and threat for environmental sustainability in Pakistan. Pakistan produces 20 million tons of solid waste annually, which has been drastically increasing 2% every year. The major metropolitan area of Pakistan is producing 71,000 tons of solid waste per day (United Nations 2018). The waste sector is producing comparatively low GHGs compared to energy and industrial sectors. The cumulative emission of GHGs in 2015 from waste sector is recorded 417.84 (Gg CO_2 equivalent), which comprises methane 237.51, carbon dioxide 0.53 and nitroxide 179.8 (Gg CO_2 equivalent), respectively (Ilmas et al. 2018).

While in the case of the land-use change and Forestry sector of Pakistan, the calculated values through the NDGM model are $\beta_1 = 1.289173$, $\beta_2 = 6{,}048{,}574.1$, $\beta_3 = 20{,}138{,}514.6$ and $\beta_4 = 20{,}593.2$. In Table 5, GHGs emissions of land use change and forestry sector have shown the increasing trend from 1990 to 2014 by utilizing the NDGM prediction model. The simulated values start 21 million ($MtCO_2e$) and reached 29 million ($MtCO_2e$). The original values of GHGs in the industrial sector are same than simulated values which can be noted in the NDGM column of the table. Whereas the forecasted values for the land use change and forestry sector in Pakistan have been calculated for the year 2015 to 2024, these predicted values are elevating from 31 million ($MtCO_2e$) and reached a level of 130 million ($MtCO_2e$).

According to World Wildlife Fund (WWF), Pakistan is the second in Asia in terms of deforestation. Due to human activities and land cover alteration, many environmental issues are arising at the local and global level. Land use is contributing 30% of GHG emission, whereas about half of GHG emission is emitted by deforestation (Szulczyk et al. 2020). The level of GHGs emission in Pakistan has increased up to 3% due to deforestation from 5 to 2.5% (United Nation Climate Change 2019).

We employed the NDGM model to forecast GHGs emissions in five major sectors of Pakistan. The findings reveal that the MAPE accuracy level for the industrial sector is slightly higher than the other sectors of Pakistan when calculated through the NDGM, while the average MAPE accuracy level for the NDGM was recorded 97.77% and is presented in Table 6.

Further, Table 7 demonstrates the ranking order of the Pakistan sector-wise for the RGR and the doubling time (D_t) based on the actual and predicted dataset. The RGR equation for actual data presents a ranking order given below:

$$\text{Industry}_{(0.170)} > \text{Energy}_{(0.157)} > \text{Agriculture}_{(0.149)} > \text{Waste}_{(0.140)} > \text{Land}_{(0.130)}$$

When we calculated the required time for the reduction of GHGs emission in Pakistan sector-wise based on the original data set, the following sequence was observed as per doubling time (D_t) model:

$$\text{Land}_{(2.990)} > \text{Waste}_{(2.960)} > \text{Agriculture}_{(2.869)} > \text{Industry}_{(2.638)} > \text{Energy}_{(2.080)}$$

The outcomes as mentioned above illustrate that the RGR of GHGs emissions in the industrial and energy sector is relatively higher than agriculture, waste, land-use change,

Table 6 MAPE % for different sectors of Pakistan

Sectors	MAPE % NDGM
Energy	2.85
Industrial processes	1.50
Agriculture	2.31
Waste	3.40
Land-use change and forestry	1.10
Average MAPE %	2.232
Overall accuracy level	97.768

Table 7 Ranking of RGR and doubling time for each sector

Relative growth rate/double time	Ranking
RGR (actual data)	$\text{Industry}_{(0.170)} > \text{Energy}_{(0.157)} > \text{Agriculture}_{(0.149)} > \text{Waste}_{(0.140)} > \text{Land}_{(0.130)}$
D_t (actual data)	$\text{Land}_{(2.990)} > \text{Waste}_{(2.960)} > \text{Agriculture}_{(2.869)} > \text{Industry}_{(2.638)} > \text{Energy}_{(2.080)}$
RGD (simulated data)	$\text{Land}_{(0.329)} > \text{Industry}_{(0.275)} > \text{Agriculture}_{(0.268)} > \text{Energy}_{(0.267)} > \text{Waste}_{(0.266)}$
D_t (simulated data)	$\text{Waste}_{(2.220)} > \text{Energy}_{(2.211)} > \text{Agriculture}_{(2.164)} > \text{Industry}_{(2.079)} > \text{Land}_{(1.870)}$

and forestry sector based on actual data set. However, these findings also show that the agricultural, industrial and energy sector requires additional time and efforts to reduce the emission of GHGs for a given RGR compared to Pakistan's waste, land use power and forestry sector.

Moreover, we used simulated data based on NDGM to assess the 2016–2024 GHG emission in Pakistan. The following results are obtained, as shown by the RGR sequence:

$$\text{Land}_{(0.329)} > \text{Industry}_{(0.275)} > \text{Agriculture}_{(0.268)} > \text{Energy}_{(0.267)} > \text{Waste}_{(0.266)}$$

A different pattern is observed based on simulated results. For the period 2016–2024, land-use change and the industrial sector may endure progressive GHGs emissions in the form of the relative growth rate of 3.29% and 2.75%, respectively, followed by the agriculture, wastes, and energy sector.

Whereas doubling time (D_t) model is also utilized based on simulated values sector-wise regarding GHGs emission comparison, the following pattern of results is obtained:

$$\text{Waste}_{(2.220)} > \text{Energy}_{(2.211)} > \text{Agriculture}_{(2.164)} > \text{Industry}_{(2.079)} > \text{Land}_{(1.870)}$$

Our findings also reveal that land-use change and industry 0.329 and 0.275 years, respectively, to double reduce GHGs emissions because of the higher RGR followed by the waste, energy, and agriculture sector.

By using the NDGM based on the original and simulated data, we make a prediction of the GHGs emission for the period from 2016 to 2024.

Presently a query arises here eventually as which country may endure maximum GHGs emissions in the long run. Therefore, to respond to this query, synthetic indices by original and forecasting values were employed, which suggests that RGR $_{\text{synth}}$ the synthetic relative growth rate has been obtained as:

$$\mathrm{RGR}_{\mathrm{synth}} = \left(\mathrm{RGR}_{\mathrm{a}}\right) + (1 - \Theta)\left(\mathrm{RGR}_{\mathrm{f}}\right)$$

where $\mathrm{RGR}_{\mathrm{a}}$ shows the relative growth derived from the original data sequence, whereas $\mathrm{RGR}_{\mathrm{f}}$ shows the relative growth derived from the original data sequence. The value of Θ is taken as 0.5. Further, D_{synth} expresses synthetic doubling time (SDT) as follows:

$$D_{\mathrm{synth}} = \Theta\left(D_{\mathrm{a}}\right) + (1 - \Theta)\left(D_{\mathrm{f}}\right)$$

where D_{a} exhibits the time required double in number to reduce GHGs emissions based on original data, whereas D_{f} presents the time required double in number to reduce GHGs emissions based on forecasting values presented in Table 8.

$$\mathrm{Land}_{(0.235)} > \mathrm{Industry}_{(0.230)} > \mathrm{Agriculture}_{(0.211)} > \mathrm{Energy}_{(0.210)} > \mathrm{Waste}_{(0.200)}$$

As per doubling time D_{t} comparison based on SDT sector-wise, we contained a sequence as follows:

$$\mathrm{Waste}_{(2.590)} > \mathrm{Agriculture}_{(2.570)} > \mathrm{Land}_{(2.430)} > \mathrm{Industry}_{(2.359)} > \mathrm{Energy}_{(2.146)}$$

According to SRGR and SDT, we got different rankings, which differs from the ranking sequence by RGR and doubling time (D_{t}) of actual data. RGR synthetic and D synthetic sequences show land-use change and the industrial sector may endure progressive GHGs emissions seen in Fig. 3. The RGR of land-use change sector was recorded higher 0.235 among all five sectors, followed by Industrial sector 0.230, agriculture sector 0.211, energy sector 0.210, and waste sector 0.200, respectively. However, the waste and agricultural sectors require more time to reduce double in numbers the GHGs emission as compared to the industrial sector in Pakistan. On the contrary, the energy sector requires relatively more time to reduce GHGs emissions as compared to other sectors.

5 Conclusion

In this study, we attempted to assess the growth and analyzed the future trends of GHGs emissions in Pakistan of five high emitting major sectors such as agriculture, industry, energy, land-use change, and forestry, and waste. To achieve this this objective, we used the GHGs emission data from 1990 to 2016. We take this process a step further and applied novel NDGM forecasting model to forecast the future trends of GHGs in Pakistan. Furthermore, we forecasted the GHG emissions until 2024 in all these sectors through NDGM and concluded that we have achieved the 97.77% overall accuracy of NDGM model. Further, we used synthetic RGR and conducted synthetic doubling

Table 8 Ranking w.r.t synthetic RGR and doubling time model

Synthetic relative growth rate/synthetic double time models	Ranking
RGR (actual data)	$\mathrm{Land}_{(0.235)} > \mathrm{Industry}_{(0.230)} > \mathrm{Agriculture}_{(0.211)} > \mathrm{Energy}_{(0.210)} > \mathrm{Waste}_{(0.200)}$
D_{t} (actual Data)	$\mathrm{Waste}_{(2.590)} > \mathrm{Agriculture}_{(2.570)} > \mathrm{Land}_{(2.430)} > \mathrm{Industry}_{(2.359)} > \mathrm{Energy}_{(2.146)}$

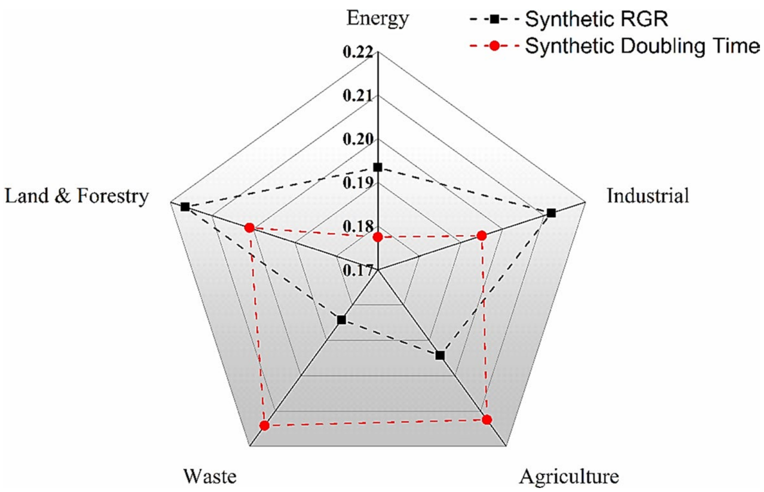

Fig. 3 Synthetic RGR and doubling time-based ranking of GHG of five sectors of Pakistan

time analysis which identified that the industrial sector and land-use change and forestry are major contributors to GHG emissions while the agriculture and waste sectors GHG emission reduction could be double in numbers as compared to industry, energy, land-use change and forestry sectors. All the five sectors showed the uprising trend of GHGs emission till 2024. The energy, waste, and agriculture sectors showed the higher growth rate of emissions among others, which implies that the energy, agriculture, and waste sectors required more serious attentions to mitigate the GHGs emission. Moreover, industrial and land-use change, and forestry sectors are also contributing but comparatively less to country GHGs emission and will rise to 36,053,420.38 and 206,111,516.6 ($MtCO_2e$) till 2024, respectively. The synthetic doubling time models showed that the waste and agriculture sectors required 2.590 and 2.570 time to decrease their GHGs emission double in number.

This means that if GHG emissions rise with the same pattern, it will be a serious threat in the future. To minimize and control air pollution, the government needs to take serious steps in order to mitigate the GHGs. In addition to identifying development areas or regions for GHGs, the government should also examine the types of GHG contaminants in these areas. New regulations and tracking networks must be established to reduce emissions as they function effectively. The Government should start awareness programs and promotions may change the public's minds toward waste management by various means (e.g., print media, digital media, and social media).

Besides, the new industry should begin by taking mitigation objectives into account and make the environment more sustainable.

Compliance with ethical standards

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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